



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Department of Artificial Intelligence

Indian Institute of Technology Hyderabad

Mid-Term Examination

AI2200: Concentration Inequalities

July-August 2026

Student Details

Name

Roll Number

12 August 2026

15 Marks

Part 01

Instructions

1. Fill in your name and roll number in the space provided above.
2. This part is for 15 marks. The complete examination is for 30 marks and counts towards 50% of the total grade.
3. Show all your working clearly. Intermediate logical steps may receive partial credit.
4. Plagiarism will not be entertained. If you are caught cheating during the examination, your answer script will not be evaluated.

1 Useful Results

1. The moment generating function of any random variable X is defined as

$$M_X(t) = \mathbb{E}[e^{tX}], \quad t \in \mathbb{R}.$$

2. For any non-negative random variable X with finite mean $\mathbb{E}[X]$,

$$\mathbb{P}\left(\{X \geq \varepsilon\}\right) \leq \frac{\mathbb{E}[X]}{\varepsilon} \quad \forall \varepsilon > 0.$$

3. For any random variable X with finite mean μ and variance σ^2 ,

$$\mathbb{P}\left(\{|X - \mu| \geq \varepsilon\}\right) \leq \frac{\sigma^2}{\varepsilon^2} \quad \forall \varepsilon > 0.$$

4. For any $\varepsilon > 0$,

$$\mathbb{P}\left(\{X \geq \varepsilon\}\right) \leq \inf_{t>0} \frac{M_X(t)}{e^{t\varepsilon}}, \quad \mathbb{P}\left(\{X \leq -\varepsilon\}\right) \leq \inf_{t<0} \frac{M_X(t)}{e^{t\varepsilon}}.$$

5. For any convex function g ,

$$\mathbb{E}[g(X)] \geq g(\mathbb{E}[X]).$$

Question 1 (2+3 Marks)

Fix $n \in \mathbb{N}$. Suppose that $X_1, \dots, X_n$ are independent (but not necessarily identically distributed), non-negative, continuous random variables. Furthermore, suppose that for each $i \in \{1, \dots, n\}$, the PDF of the random variable X_i satisfies the condition

$$f_{X_i}(x) \leq 1 \quad \forall x \geq 0.$$

(a) Prove that for each $i \in \{1, \dots, n\}$,

$$\mathbb{E}[e^{-tX_i}] \leq \frac{1}{t} \quad \forall t > 0.$$

(b) Prove that for all $\varepsilon > 0$,

$$\mathbb{P}\left(\left\{\sum_{k=1}^n X_k \leq n\varepsilon\right\}\right) \leq (e\varepsilon)^n.$$

Question 2 (2+3 Marks)

Fix $n \in \mathbb{N}$, and suppose that $X_1, \dots, X_n$ are independent and identically distributed with a finite mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$. Let $S_n = \sum_{k=1}^n X_k$. We will show in two distinct ways that

$$\mathbb{E} \left[\left| \frac{S_n}{n} - \mu \right| \right] = O \left(\frac{1}{\sqrt{n}} \right).$$

(a) Using Jensen's inequality, show that

$$\mathbb{E} \left[\left| \frac{S_n}{n} - \mu \right| \right] \leq \frac{\sigma}{\sqrt{n}}.$$

Clearly indicate the function to which you apply Jensen's inequality.

(b) It is a known result in probability theory that

$$\mathbb{E} \left[\left| \frac{S_n}{n} - \mu \right| \right] = \int_0^\infty \mathbb{P} \left(\left| \frac{S_n}{n} - \mu \right| > \varepsilon \right) d\varepsilon.$$

Taking the above formula on faith, express the integral on the right-hand side as a sum of two integrals, one with limits from 0 to a and another with limits from a to ∞ , where $a > 0$. Upper bound each of the integrals carefully, and optimise over $a > 0$ to eventually show that

$$\mathbb{E} \left[\left| \frac{S_n}{n} - \mu \right| \right] \leq \frac{2\sigma}{\sqrt{n}}.$$

Question 3 (4+1 Marks)

Consider the following hypotheses is true about the PDF f of a continuous random variable X :

$$\begin{aligned}\mathcal{H}_0 : \quad f(x) = f_0(x) &= \begin{cases} 1, & x \in (0, 1), \\ 0, & \text{otherwise,} \end{cases} \\ \mathcal{H}_1 : \quad f(x) = f_1(x) &= \begin{cases} \frac{2}{3}(x+1), & x \in (0, 1), \\ 0, & \text{otherwise.} \end{cases}\end{aligned}$$

A classifier $T : (0, 1) \rightarrow \{\mathcal{H}_0, \mathcal{H}_1\}$ is constructed as follows:

$$T(x) = \begin{cases} \mathcal{H}_1, & \frac{f_1(x)}{f_0(x)} \geq \tau, \\ \mathcal{H}_0, & \text{otherwise,} \end{cases}$$

where $\tau > 0$ is an adjustable hyperparameter of the classifier. The risk of classifier T is defined as

$$R(T) = \frac{2}{3} \mathbb{P}_{\mathcal{H}_0}(\{T(X) = \mathcal{H}_1\}) + \frac{1}{3} \mathbb{P}_{\mathcal{H}_1}(\{T(X) = \mathcal{H}_0\}),$$

where $\mathbb{P}_{\mathcal{H}_0}$ (respectively, $\mathbb{P}_{\mathcal{H}_1}$) denotes probability computed under the assumption that $\mathcal{H}_0$ (respectively, $\mathcal{H}_1$) were true.

- (a) Determine the value of τ for which the risk $R(T)$ is minimised.
- (b) Compute the optimal risk obtained by using the value of τ determined in part (a).



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Part 02

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2 Useful Results

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5. For any convex function g ,

$$\mathbb{E}[g(X)] \geq g(\mathbb{E}[X]).$$

Question 1 (4+2+4 Marks)

For a fixed $0 < b < +\infty$, suppose that X has the PDF

$$f_X(x) = \frac{1}{2b} e^{-|x|/b}, \quad x \in \mathbb{R}.$$

- (a) Compute $\mathbb{E}[X]$ and $\text{Var}(X)$.
- (b) Compute the moment generating function of X .
- (c) Use Chernoff's technique to derive a bound on $\mathbb{P}(\{X \geq \varepsilon\})$ for any $\varepsilon > 0$.

Question 2 (1+2+2 Marks)

Given any $A \subseteq \mathbb{R}$, let a binary classifier g_A be defined via

$$g_A : \mathbb{R} \rightarrow \{0, 1\}, \quad g_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Furthermore, let $\mathcal{G}$ be the uncountably infinite collection of classifiers given by

$$\mathcal{G} = \{g_A : A \subseteq \mathbb{R}\}.$$

Consider n IID data points $(X_1, Y_1), \dots, (X_n, Y_n)$ in which

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} \text{Ber}(0.8), \quad X_i \text{ is independent of } Y_i \text{ for each } i.$$

(a) For $A = [0, +\infty)$, evaluate the risk

$$R(g_A) = \mathbb{P}(\{g_A(X) \neq Y\}).$$

(b) Given a fixed $A \subseteq \mathbb{R}$, define the empirical risk of classifier g_A as

$$\hat{R}_n(g_A) = \frac{1}{n} \sum_{k=1}^n \mathbf{1}_{\{g_A(X_k) \neq Y_k\}} = \frac{1}{n} \sum_{k=1}^n \left(\mathbf{1}_{\{g_A(X_k)=0, Y_k=1\}} + \mathbf{1}_{\{g_A(X_k)=1, Y_k=0\}} \right).$$

For what choice of A does the corresponding classifier g_A have minimal empirical risk?

Hint: How can you make the sum of indicators inside the brackets of the last expression zero?

(c) For the empirical risk minimiser identified in part (b), say $\hat{g}_n$, evaluate $R(\hat{g}_n)$ and $\hat{R}_n(\hat{g}_n)$.